

Appendix: Artifact Evaluation

Artifact Evaluation (AE)

Paper ID: pap142

This appendix evaluates the three computational artifacts defined in the AD:

- A1: TempGNN reference model, tests, and edge-stream profiling.
 - A2: four-system U280 forward-path execution and measurement workflow.
 - A3: source-labeled paper-reference CSV/SVG reconstruction.
- The recommended reviewer order is A1 → A3 → A2.

The first two paths do not

require FPGA access. Artifact A2 uses an author-provided Alveo U280 account;

credentials are delivered only through the private SC submission channel.

I. EVALUATION SCOPE AND BADGE BOUNDARY

The artifact is prepared to support Artifacts Available and Artifacts

Evaluated-Functional. Apache-2.0 is included; a version-specific DOI must be

present by the August 25, 2026 artifact freeze.

The current package does not assert Results Reproduced. The fresh U280 path

uses bounded 8,192-event prefixes, 8-dimensional Q10 kernels, deterministic

stand-in weights, and xbutil total-board power. The paper uses complete model

configurations, default 32-bit floating point, trained checkpoints, full

evaluation streams, and post-route Vivado power estimates. The configuration

therefore records `results_reproduced_eligible: false`. Packaged reference rows

are never substituted for a failed fresh measurement.

Expected direct review time, excluding optional Vitis rebuilds:

Artifact	Setup, execution, and analysis time
A1	5-10 min setup; less than 2 min execution; less than 2 min analysis
A2	10-20 min setup; 30-90 min execution; less than 5 min analysis
A3	less than 5 min setup; less than 1 min execution; 2-5 min analysis

II. COMPUTATIONAL ARTIFACT A1

A. Artifact Setup

Use Linux x86_64 with Python 3.10 or newer, GNU Make, and Bash:

```
git clone https://github.com/TempGNN-Program/
→ TempGNN.git
cd TempGNN
python3 -m venv .venv
```

```
source .venv/bin/activate
pip install -r requirements.txt
```

The frozen package may be used instead of `git clone`:
`tar -xzf pap142_tempgnn_sc26_ae_u280.tgz`
`cd pap142_tempgnn_sc26_ae`

No FPGA is needed. Unit fixtures are generated locally. Optional Q14 profiling downloads TGL edge streams; packaged U280-prefix samples are already under `external/u280_dataset_samples/` with URL and SHA256 metadata.

B. Artifact Execution

The task dependency chain is:

```
A1-T1 environment
→ A1-T2 unit tests for TDP/DDTC/OATS and
→ artifact contracts
→ A1-T3 optional real-edge acquisition
→ A1-T4 optional edge-stream profiling
→ A1-T5 CSV/Markdown summary generation
```

Run the required fast path:

```
python3 -m unittest discover -s tests
```

Expected terminal summary:

```
Ran 28 tests
OK
```

Run the optional real-edge path:

```
make data
make q14
```

The default Q14 parameters are WIKI/MOOC/REDDIT, four models, fanout 20, depth

2, and the batch parameters recorded in each output row. `make data` is skipped

when the corresponding edge files are already present.

C. Artifact Analysis

All 28 tests must pass. In particular, the recursive TDP state must agree with

chronological updates, fixtures must be deterministic, xclbin link requests

must agree with the Vivado-connected clock and nonnegative post-route WNS/TNS,

and duplicate xclbins must be rejected by preflight.

Optional Q14 outputs:

```
results/q14_real_tgl_edges/q14_dataset_model_
→ summary.csv
results/q14_real_tgl_edges/q14_batches.csv
results/q14_real_tgl_edges/q14_summary.md
```

Inspect packet hit/reuse rates, collisions, stalls, and memory traffic. The

latency columns in this CPU path use a 225 MHz cycle model and are not fresh

U280 board timings. Passing A1 demonstrates that the mechanisms supporting

contributions C1-C3 are executable and internally checked.

III. COMPUTATIONAL ARTIFACT A2

A. Artifact Setup

Hardware and recorded software environment:

```
Board: AMD/Xilinx Alveo U280,  
↳ xcu280-fsvh2892-2L-e  
Platform: xilinx_u280_gen3x16_xdma_1_202211_1  
OS: Ubuntu 22.04 x86_64  
XRT: 2.16.204  
Vitis/Vivado: 2023.2 (rebuild only)  
GCC: 11.4.0  
GNU Make: 4.3  
Python: 3.10 or newer
```

The reviewer receives a dedicated, time-bounded account through the SC private

channel. After login:

```
source /opt/xilinx/xrt/setup.sh  
cd /path/to/pap142_tempgnn_sc26_ae  
xrt-smi examine
```

The four metadata-normalized artifacts are:

```
artifacts/u280/TempGNN/bin/tempgnn_forward_ke_   
↳ rnel.hw.xclbin  
artifacts/u280/MATG/bin/matg_kernel.hw.xclbin  
artifacts/u280/ViTeGNN/bin/vitegnn_kernel.hw.   
↳ xclbin  
artifacts/u280/RTGA/bin/rtga_kernel.hw.xclbin
```

The common XRT host and baseline C-sim reference binaries are colocated under

the same bin/ directories. Source, paper-mechanism mapping, and limitations

are in hardware/baselines/README.md and hardware/baselines/MECHANISM_MAP.md.

The repository and frozen package also include the exact real-input fixture

metadata and goldens used by the packaged run under results/generated_u280_comparison_fixtures/.

Optional rebuilds require:

```
source  
↳ /tools/Xilinx/Vitis/2023.2/settings64.sh
```

Rebuilding is not required for evaluation and can take several hours per

xclbin. The direct-run path stays within the SC26 review budget.

B. Artifact Execution

The workflow is implemented by scripts/run_u280_core_reproduction.py:

```
A2-T1 preflight four source revisions, hosts,  
↳ and distinct xclbin hashes  
-> A2-T2 generate fixtures from bundled  
↳ real-data prefixes  
-> A2-T3 generate independent software  
↳ goldens  
-> A2-T4 load and run each implementation on  
↳ U280  
-> A2-T5 gate the repeated-kernel window and  
↳ sample total board power  
-> A2-T6 validate checksums, repetitions,  
↳ energy, provenance, and clocks  
-> A2-T7 aggregate raw rows and derive  
↳ diagnostic figure-shaped tables
```

-> A2-T8 compare fresh tables with the
↳ paper-reference rows

Run preflight:

```
make u280-core-preflight
```

Expected: U280 four-implementation preflight PASS, followed by four different

xclbin SHA256 values.

Run the complete direct measurement:

```
make ae-core-u280 U280_CORE_DEVICE=0  
↳ U280_CORE_REPETITIONS=3
```

Checked parameters are loaded from configs/u280_core_reproduction.json:

```
datasets: WK, MC, RT, LM, WT, GT  
models: JODIE, TGN, TGAT, APAN  
implementations: TempGNN, MATG, ViTeGNN, RTGA  
real prefix: 8,192 events per dataset  
target batch: 1,000  
repetitions: 3  
tensor path: 8-dimensional Q10  
requested clock: 225 MHz  
achieved-clock comparison tolerance: 0.5 MHz
```

Each dataset/model pair is calibrated to a repeated-kernel measurement window.

The host performs a warmup, waits at a file gate, and then executes the measured

iterations while the harness samples xbutil examine --report electrical.

Optional one-fixture TempGNN sanity command:

```
make u280-run U280_DEVICE=0
```

Optional rebuild commands:

```
make u280-build \  
U280_PLATFORM=/path/to/xilinx_u280_gen3x16_   
↳ xdma_1_202211_1.xpfm  
make u280-baseline-build  
make u280-stage-artifacts
```

Baseline links should be run serially on a 64 GB host. The staging command

redacts login, host, and absolute paths from textual evidence and the xclbin

BUILD_METADATA/SYSTEM_METADATA sections; it verifies that the FPGA

BITSTREAM section is byte identical before and after normalization.

C. Artifact Analysis

The run creates a new directory and never overwrites paper-reference data:

```
results/reviewer_u280_runs/<run-id>/provenanc_   
↳ e.json  
results/reviewer_u280_runs/<run-id>/raw/TempG_   
↳ NN/measurements.csv  
results/reviewer_u280_runs/<run-id>/raw/MATG/_   
↳ measurements.csv  
results/reviewer_u280_runs/<run-id>/raw/ViTeG_   
↳ NN/measurements.csv  
results/reviewer_u280_runs/<run-id>/raw/RTGA/_   
↳ measurements.csv  
results/reviewer_u280_runs/<run-id>/baselines_   
↳ _u280/  
results/reviewer_u280_runs/<run-id>/derived_c_   
↳ omparison_figures/
```

```
results/reviewer_u280_runs/<run-id>/verificat_
↪ ion.json
results/reviewer_u280_runs/<run-id>/verificat_
↪ ion.md
```

The packaged final run is

```
results/reviewer_u280_runs/pap142_u280_
measured_20260715T052052Z/; its four
raw CSV files contain 72 rows each (288 total). The packaged
one-repetition
smoke run is pap142_u280_smoke_
20260715T052052Z.
```

For 3 repetitions, each implementation must contain 72 rows and the combined

matrix must contain 288 rows. Every row must report:

- golden_validation=PASS and repeat_consistency=PASS;
- a nonzero kernel and embedding checksum;
- a real dataset source URL and 64-character input/golden hashes;
- positive latency and power;
- energy consistent with $\text{latency_ms} * \text{power_w}$;
- requested and xclbin-link-requested clocks, the Vivado-connected post-route kernel clock, WNS/TNS, and timing_met=PASS.

The workflow refuses normalized comparison generation if an implementation's

post-route clock changes between rows or if the four timing-closed clocks

differ by more than 0.5 MHz. verification.md reports the maximum relative error against

paper-reference Fig.11 and the U280 subset of Fig.12. A diagnostic tolerance

FAIL is a valid completed run and must remain visible. It does not demonstrate

paper equivalence and must not be converted to PASS by adjusting measured rows.

Passing A2 establishes that four distinct implementations execute on the

same U280, produce stable software-checked outputs, and emit traceable raw

measurement data. The bounded reproduction scope prevents this path from independently

supporting a Results Reproduced claim for C4.

IV. COMPUTATIONAL ARTIFACT A3

A. Artifact Setup

Use the Python environment created for A1. No FPGA, GPU, external dataset, or

network access is required. Inputs are:

```
reference_inputs/paper_figure_values.csv
reference_inputs/README.md
```

The table has 668 rows. Every row records its source class, locator, and

uncertainty. Original author workbooks and vector files are not distributed

because they contain author metadata; only the anonymized numeric extraction

and source hashes are packaged.

B. Artifact Execution

The dependency chain is:

```
A3-T1 validate the reference table and source
↪ labels
-> A3-T2 split records by figure
-> A3-T3 generate matching CSV and SVG files
-> A3-T4 generate the combined table and
↪ manifest
-> A3-T5 run consistency tests
-> A3-T6 regenerate the reviewer report
```

Commands:

```
python3 -m scripts.reproduce_paper_figures
python3 -m scripts.make_ae_report \
--q14-summary results/q14_real_tgl_edges/q1_
↪ 4_dataset_model_summary.csv \
--board-json results/board_u280/summary.json
↪ \
--out results/ae_report
```

Equivalent Make target:

```
make smoke
make report

make smoke includes the 28 tests from A1.
```

C. Artifact Analysis

Expected outputs:

```
results/paper_reproduction/all_figure_data.csv
results/paper_reproduction/figure_data_manife_
↪ st.csv
results/paper_reproduction/fig2_execution_bre_
↪ akdown.csv/.svg
results/paper_reproduction/fig4a_branch_paral_
↪ lelism_ratio.csv/.svg
results/paper_reproduction/fig9b_gpu_overhead_
↪ _breakdown.csv/.svg
results/paper_reproduction/fig10_speedup_tgli_
↪ te_cpu.csv/.svg
results/paper_reproduction/fig11_speedup_matg_
↪ .csv/.svg
results/paper_reproduction/fig12_energy_tempg_
↪ nn.csv/.svg
results/paper_reproduction/fig13_ablation_tim_
↪ e.csv/.svg
results/paper_reproduction/fig14a_batch_sensi_
↪ tivity.csv/.svg
results/paper_reproduction/fig14b_tdp_entries_
↪ .csv/.svg
```

Expected summary checks:

```
TempGNN vs TGLite-CPU: 132.80x
TempGNN-G vs TGLite-CPU: 10.85x
Cascade vs TGLite-CPU: 5.22x
TempGNN vs MATG, explicit plotted AVG: 7.7889x
Paper prose for TempGNN vs MATG: 7.6x
Energy, Cascade/TempGNN, explicit plotted AVG:
↪ 33.545x
w/o DDTC normalized time: 3.08x
w/o OATS normalized time: 1.77x
```

The Fig.11 explicit AVG and the paper's rounded prose value are both preserved.

The generator must not alter individual values to force agreement. Generated

SVG titles contain Paper-reference, and the manifest identifies whether each series came from exact workbook cells or axis-calibrated vector geometry.

Passing A3 demonstrates deterministic reconstruction and provenance of the packaged paper plotting records for C4. It does not demonstrate a fresh rerun of MATG, ViTeGNN, RTGA, Cascade, TGLite-CPU, or the complete TempGNN paper configuration.

V. BADGE EVIDENCE SUMMARY

Badge		Current evidence
Artifacts Available		Source, tests, inputs, CSV/SVG records, four metadata-normalized U280 xclbins, reports, and documentation; Apache-2.0 included and DOI required by freeze
Artifacts Functional	Evaluated-	A1 tests, baseline C-sim, four-xclbin pre-flight, board goldens, repeated outputs, power/latency rows, and post-route evidence
Results Reproduced		Not currently asserted because A2 is a bounded mechanism-level comparison rather than the paper’s full precision, checkpoints, streams, and power method